Data Spaces Symposium

Accelerating adoption. Increasing impact.

Interoperability & standards – the key factor for scaling data spaces

State-of-play of standards and protocols
for global interoperability

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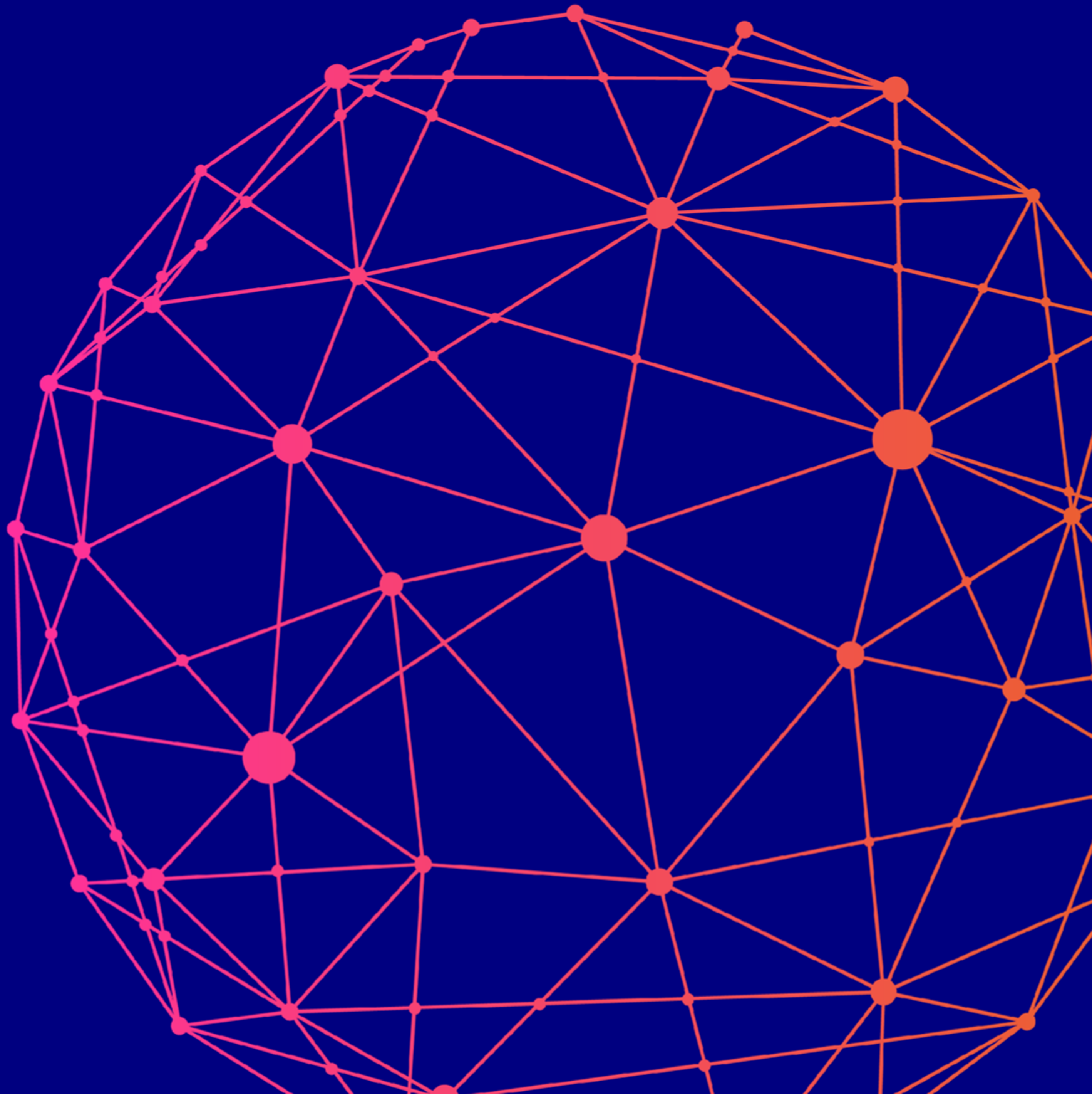
2026-02-11



A decade of pioneering.

A future of impact.

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10 years

Overview on Data Space Standards

Ongoing development in ISO/IEC JTC1 (tl;dr)

20151 multipart series (SC38):

- ISO/IEC DIS 20151-1 Dataspace concepts and characteristics (expected publication in 2026)
- ISO/IEC AWI TR 25850 (20151-3) Use cases for dataspaces (expected publication in 2026)
- ISO/IEC PWI 20151-2 (26189) Dataspace Trust Frameworks (NWIP expected soon)

- Eclipse Data Space Working Group PAS Submission:
- ISO/IEC DIS 26450 Eclipse Dataspace Protocol (DSP)
- ISO/IEC DIS 26451 Eclipse Decentralized Claims Protocol (DCP

- Also relevant (SC38):
- ISO/IEC PWI Switching by design
- ISO/IEC PWI 38011 Observability

Overview on Data Space Standards

Ongoing development in Europe (tl;dr) - 1

Harmonized European Norms in CEN/CLC JTC25

- EN 18235-1:2026 Trusted Data Transactions part 1 (in publication)
- prEN 18235-2 Trusted Data Transactions part 2 (in Enquiry)
- prEN 18235-3 Trusted Data Transactions part 3 (preparation for Enquiry)

CEN/CLC JTC 25

- Cloud switching and interoperability in a European context
- PWI Data quality in data spaces
- PWI Data Space Trust Frameworks in a EU context
- PWI Overview and architecture of standards in support of the European Trusted Data Framework
- Quality assessment of internal data governance processes
- FprCEN/CLC/TS 18331 Maturity assessment of Common European Data Spaces

- Cloud switching and interoperability in a European context

Overview on Data Space Standards

Ongoing development in Europe (tl;dr) - 2

ETSI

- Data catalogue (DCAT) profiles and extensions
- Implementation framework for semantic assets

Achievements so far

Where do we stand?

1. Alignment between International and European Standards
- Definitions
- European perspective complements international standards – and vice versa

2. Foundational international Standards finalizing
- 20151
- Trusted Data Transaction

3. European standards for Data Act §33 progressing fast
- First EN, and first TS finalizing

4. New European and international Standards upcoming.
- Data Space Trust Frameworks
- Data Quality

Data Space

Dataspace

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» **environment enabling trusted data sharing between participating parties, based on an agreed governance framework, along with an agreed set of policies, semantic models, standardised protocols, processes, and facilitating services**

Data Space

Dataspace

» **environment enabling trusted data sharing between participating parties, based on an agreed governance framework, along with an agreed set of policies, semantic models, standardised protocols, processes, and facilitating services**

» **Where a data space participant is acting in a given role.**

» **Where a data space participant role is a set of activities within a data space for the purpose of data sharing or related activities.**

Facets of a dataspace

Three primary facets, each serving distinct functions

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- **Legal and governance facet**
- enforces rights, obligations, and regulatory compliance across participants.
- Roles: Data Rights Holders (EU), Data Subject (EU), Data Intermediation Services (EU), Data Users, Data Recipients

- **Economic facet**
- manages the services, interactions, and workflows that enable value generation and marketplace activity. Notably, terms for this facet are also **Business** or **Operational Facet**.
- Roles: Data Providers, Data Consumers, Observers, Intermediaries, Marketplaces, Individuals, Service Providers

- **Technical facet**
- encompasses the architecture and protocols that facilitate secure and interoperable data exchanges.
- Roles: Data Space Participants

Trusted Data Sharing

In Data Spaces

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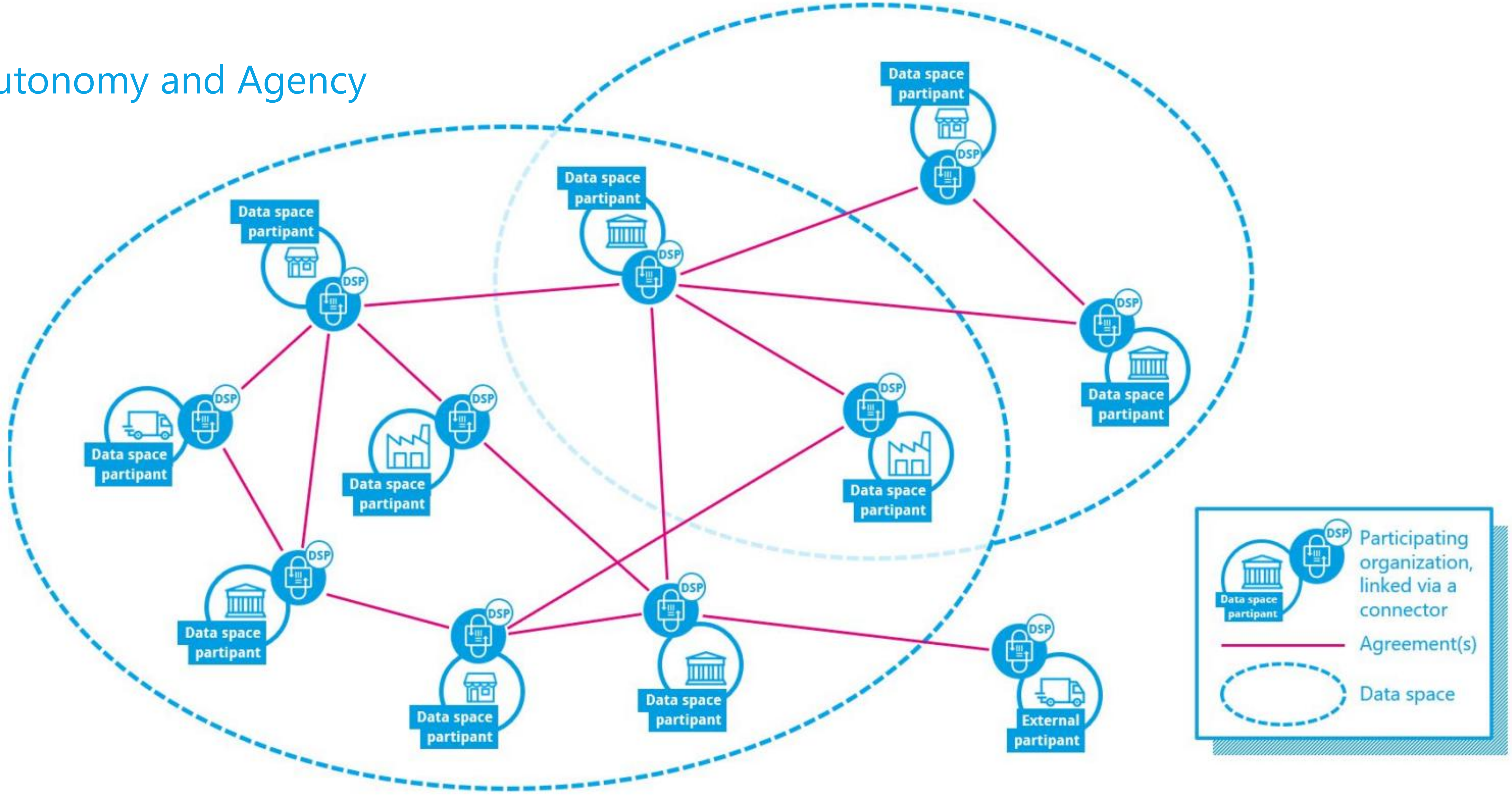
Enabling Organizational Autonomy and Agency

Facilitating interoperability

Creating resilient
and dynamic
Data Value Chains



Picture source: IDSA Data Space Connector Report

Data space characteristics | ISO/IEC 20151

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- Maintain control
- Establish trust
- Discover data
- Negotiate data sharing contracts
- Orchestration of data sharing
- Observability of action
- Interoperability

Functional components

- Multi-level policies
- Semantic models
- Communication protocols
- Processes and Rules

Interoperability

Is a multifaceted problem

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ISO/IEC 19941
Cloud Computing Interoperability and Portability

- Policy
- Behavior
- Semantic
- Syntactic
- Transport

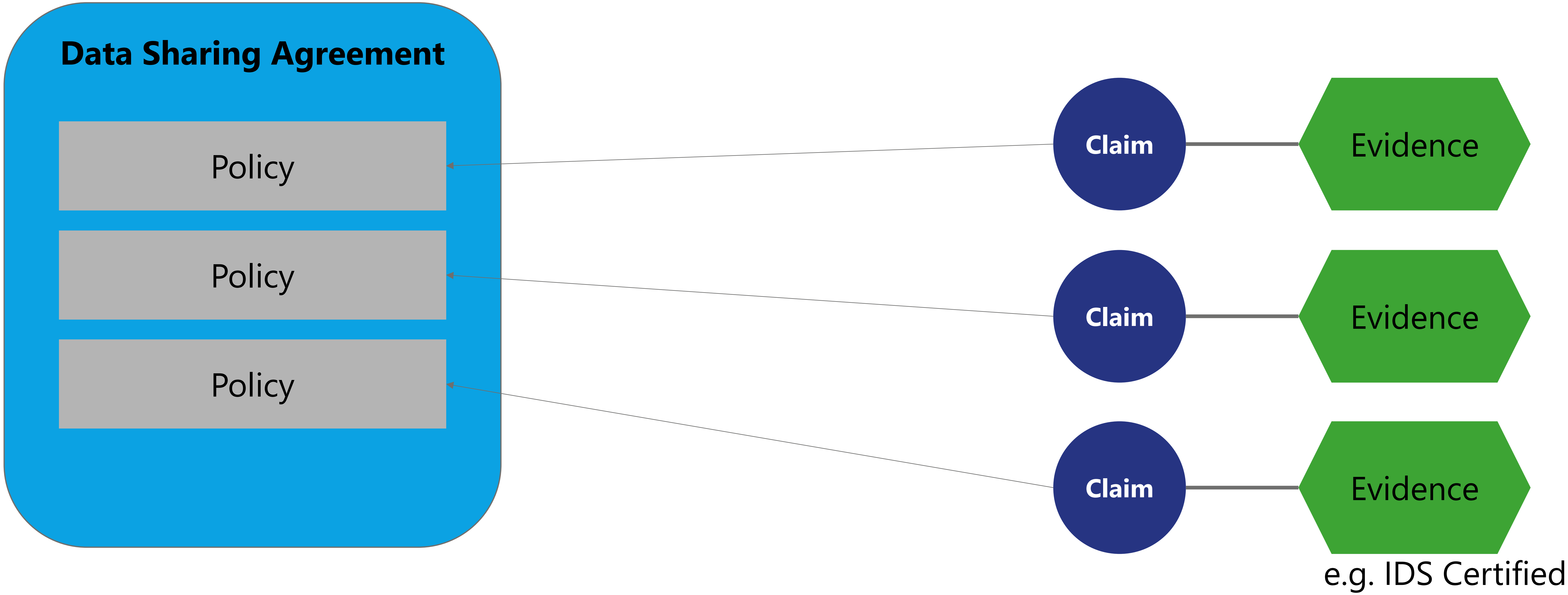
European Interoperability Framework -EIF

- Legal
- Organizational
- Semantic
- Technical

Technical Facet

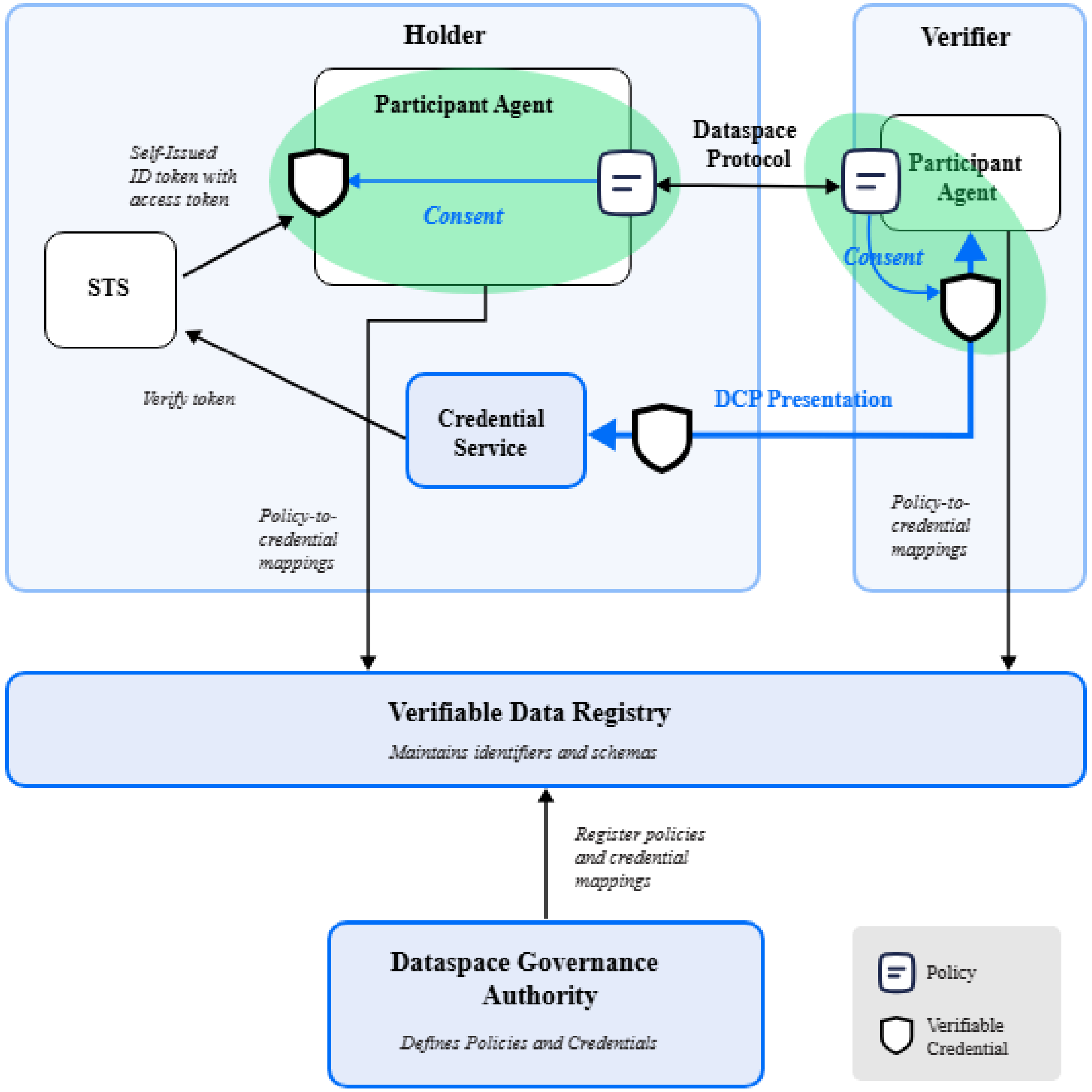
Policy and Claim Reconciliation

The automated execution of data sharing agreements



The Decentralized Claims Protocol

ISO/IEC DIS 26451



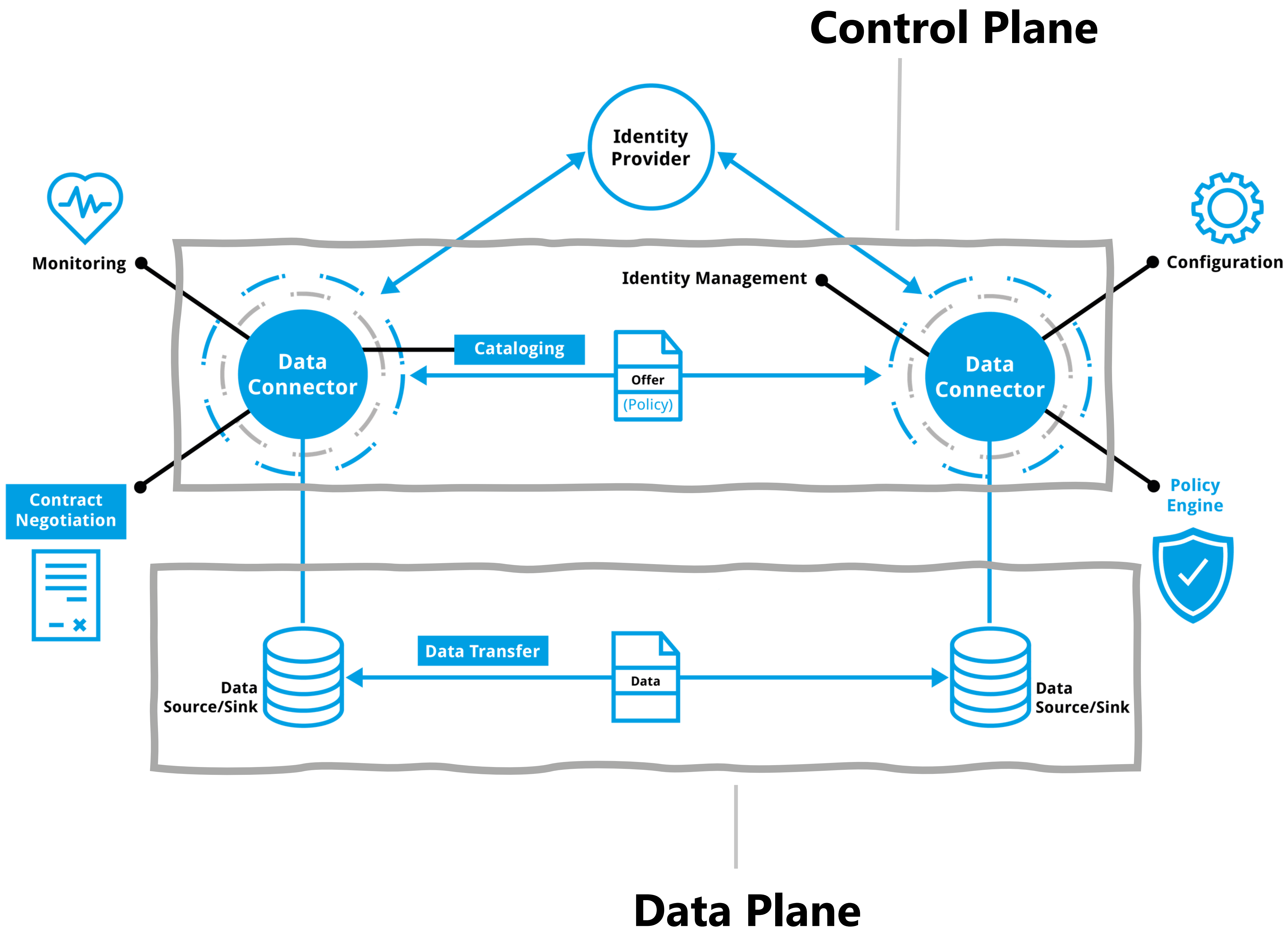
Consent is the process by which the **Holder approves the release of a Verifiable Presentation,** and the **Verifier agrees** to allow access to a protected resource.

The Dataspace Protocol is built on **machine-to-machine message exchange** patterns, presentation protocols that rely on end-user (i.e., human) consent are not applicable.

Consent is handled by first mapping Dataspace Protocol Offer and Agreement policies to a set of REQUIRED Verifiable Credentials. ...

The need for Dataspace Protocol

ISO/IEC DIS 26450



DCAT enables semantic interoperability by providing a **shared**, machine-readable vocabulary that ensures datasets are **described consistently** across organizations and domains.

ODRL enables semantic interoperability by providing a standardized, machine-readable language for expressing **usage rights and policies,** ensuring that data access and permissions are understood consistently across systems and organizations.

Aspects

Semantic interoperability of

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- Metadata, e.g catalogues
- Policies & Claims
- Data and Data Products

2/16/2026

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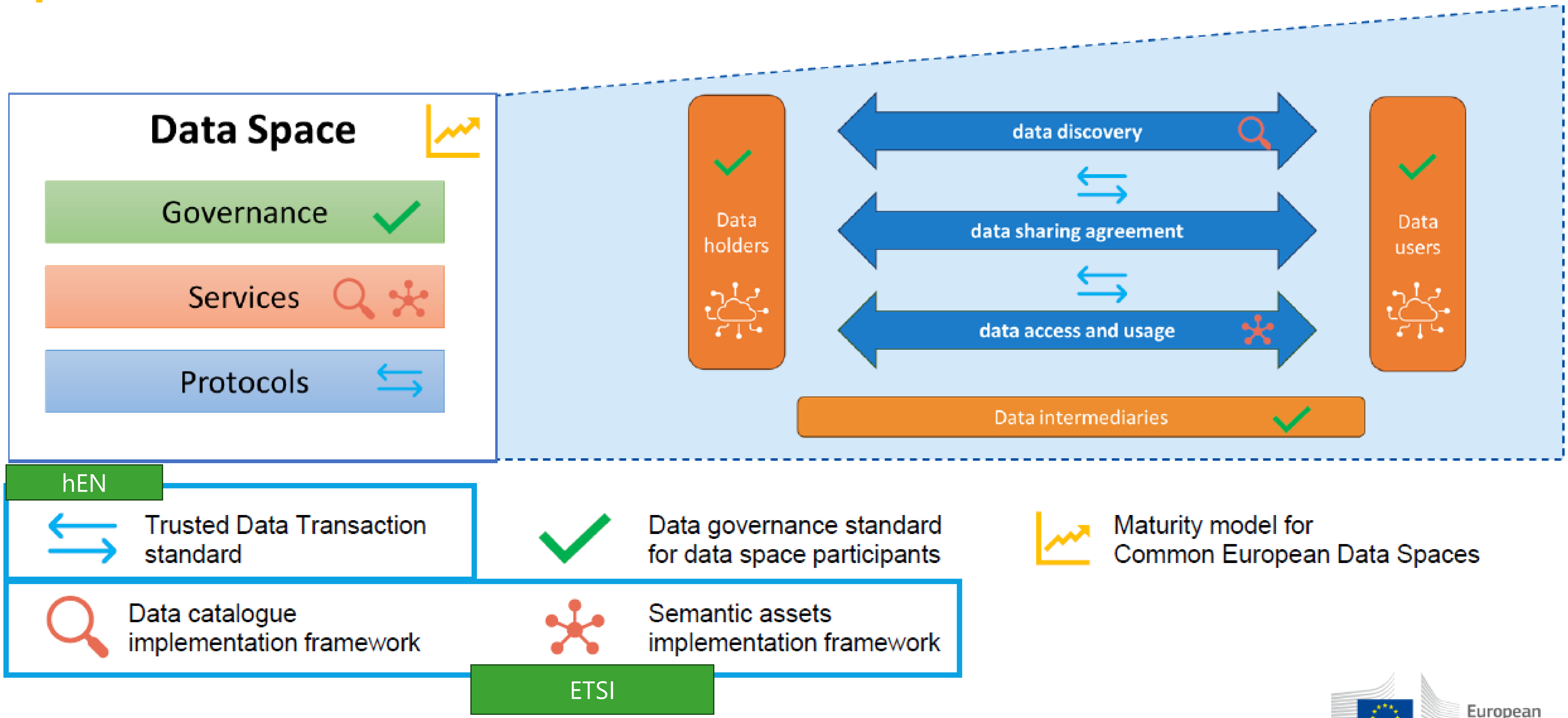
Compliance with the EU Data Regulation

Data Act §33

Standardisation request
European Trusted Data Framework



Key result Trusted Data Transaction

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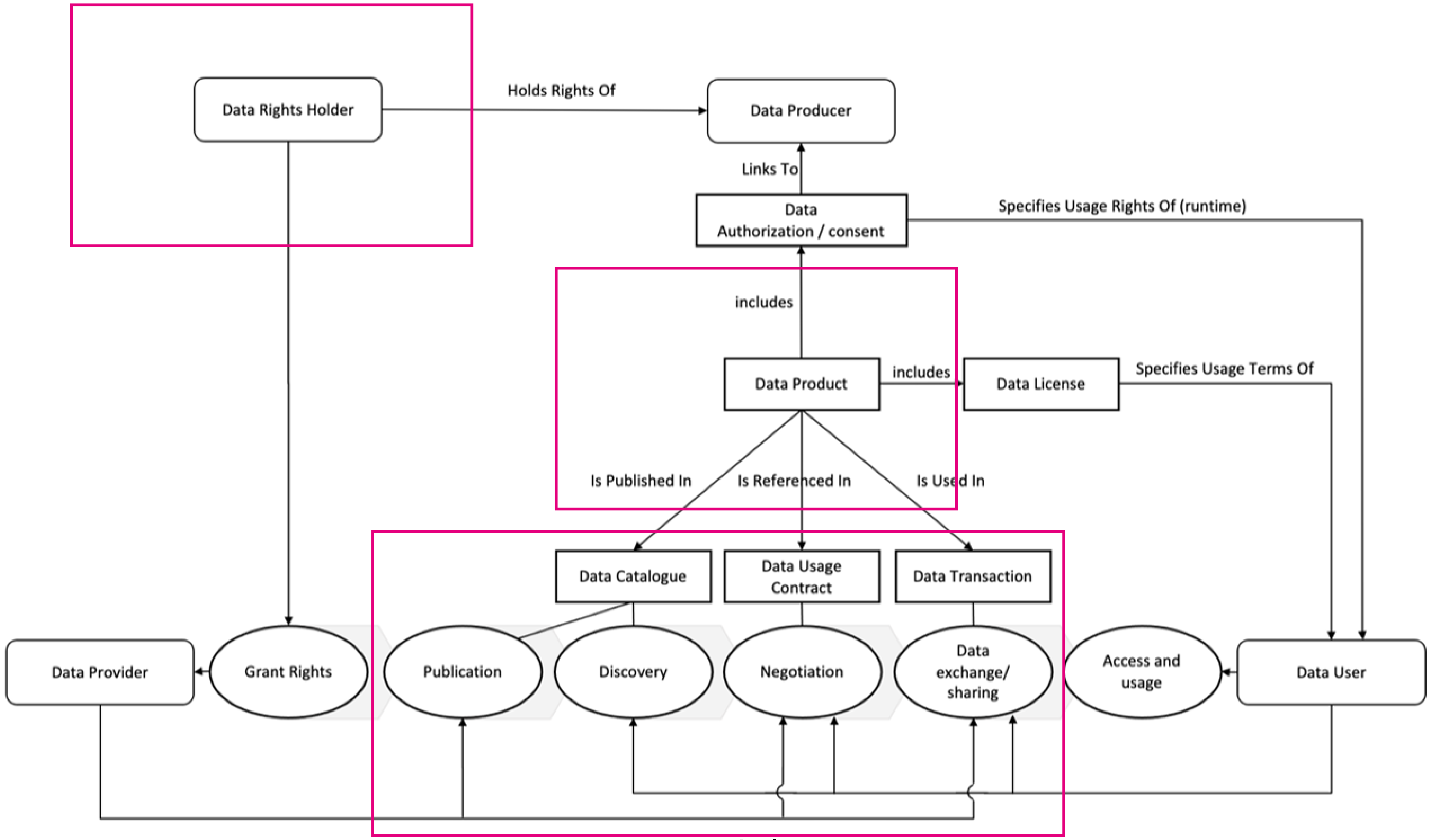

Modified from: https://www.cencenelec.eu/media/CEN-CENELEC/CWAs/RI/2024/cwa18125_2024.pdf

Trusted Data Transaction is a multipart harmonized standard to fulfill the requirements of the EU Data Act §33

Part 1: Concepts, terminology, and mechanisms,

Part 2: Trustworthiness requirements

Part 3: Interoperability requirements

Data Product in JTC 25

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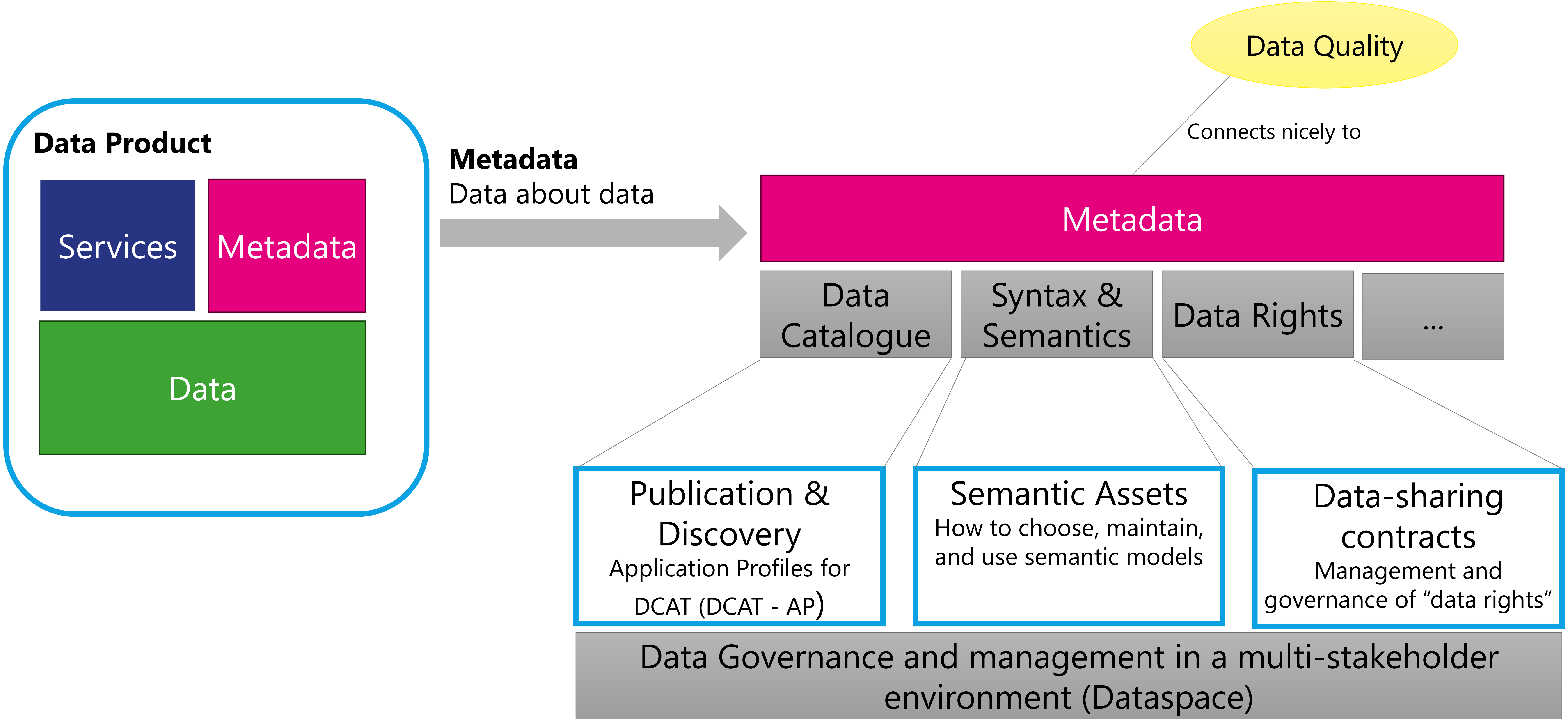

What’s next?

Upcoming work

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International

- Use Cases for dataspaces – understanding the economic facet
- Dataspace Trust Frameworks – understanding trustworthiness

Europe

- Finalizing Trusted Data Framework – hEN for Data Act Compliance
- Data Space Trust Frameworks – the European Perspective
- Internationalization of European Standards – International acceptance
- Data Quality aspects in Data Spaces – Standardisation Request expected

And Dataspace Standards in more regions of the world


10 years